EXPERIMENT 13

Building Approval-Based Workflow Automation in a Zoho Creator Application

Subject: No-Code Application Development

Aim

To model and configure a leave approval workflow in Zoho Creator using status transitions, conditional actions and notifications.

Objectives

- Define request submission and manager approval stages.
- Specify conditions for approval, rejection and notification.
- Test successful and exceptional workflow paths.

Selected Application

Leave Management app. Each request starts as Pending and can be updated by an authorized manager to Approved or Rejected.

Requirements / Tools

- Leave Request form, status field and manager role
- Appropriate Zoho Creator workflow/approval features available to the account
- Fictional employee and manager test users

No-Code Connection

A visual approval workflow consists of a trigger, condition, action and defined outcome. No-code automation is reliable only when permissions, error paths and status transitions are considered.

Illustrative Business Scenario

Set up the conceptual path Submit → Pending → Manager Review → Approved/Rejected and validate unauthorized attempts.

Experimental Procedure

Follow these steps in the designated simulator, browser or visual app builder. Menu locations and available features may vary by product version and account plan.

1. Define the business rule that a newly submitted request begins in Pending status.
2. Identify the manager responsible for reviewing a request and restrict approval actions to that role.
3. Configure an on-submit action to record the request and optionally notify the reviewer.
4. Set up approval and rejection outcomes using supported workflow/approval features.
5. Notify the employee of the decision without revealing other users’ records.
6. Test each branch and ensure a processed request is not approved a second time.

No-Code Application Perspective

- Identify the user action and the visual interface element involved (form, page, workflow or report).
- Trace the data movement from input to processing to stored record, report or notification.
- Identify configuration that replaces traditional hand-written code and note where scripting might still be required.
- Check whether the experiment can be repeated using sample records and documented settings.

Evidence to Collect

- Screenshot of the configured screen, topology, tool or application.
- A short observation table showing input, action and output.
- One successful test and one failed or exceptional test with expected behavior.

Workflow transitions

Current state	Action	Next state
New request	Successful submission	Pending
Pending	Authorized manager approves	Approved
Pending	Authorized manager rejects	Rejected
Approved/Rejected	Repeated decision attempt	Disallowed or reviewed

Test scenarios

Scenario	Expected result
Employee submits valid request	Pending record created
Manager approves request	Status becomes Approved
Manager rejects request	Status becomes Rejected
Employee attempts manager action	Access denied
Notification fails	Status remains traceable; failure logged if supported

Interpretation / Analysis

- An explicit state model prevents ambiguous or conflicting approval outcomes.
- Actual workflow configuration may vary by Zoho Creator plan and version.
- Role-based permissions and audit history are as important as convenience notifications.

Evaluation Against a No-Code Solution

- Ease of creation: determine how much is configured visually rather than programmed.
- Data and integration: identify where records, connections or services are created and managed.
- Governance: consider role permissions, traceability and safe data handling.
- Limitations: identify any feature that may require paid access, external services or custom logic.

Verification / Test Cases

Suggested Test Record

Test	Input / action	Expected observation
Normal test	Valid sample values and supported action	Expected successful output
Boundary test	Missing value, invalid input or empty data	Validation or clear empty state
Access test	Unapproved user attempts restricted action	Access limited according to role
Repeatability	Re-run same valid example	Consistent output or explain differences

Observation Notes

During the practical session, replace the illustrative outcomes with actual observed results. Include the software/platform name, version if known, screenshots, and any errors encountered.

Results

An approval workflow was specified with state transitions and acceptance tests. Live notification and role behavior must be checked after configuration.

Conclusion

Workflow automation can reduce delays and improve traceability when authorization checks and exception handling are properly designed.

No-Code Learning Outcome

This experiment links the technical concept to a practical no-code application development workflow. It highlights visual configuration, user experience, data handling and safe validation rather than treating the exercise as conventional full-code software development.

Precautions

- Use only fictional sample records for classroom demonstrations; avoid exposing passwords, tokens, employee information or real contact details.
- Record your own screenshots, observed outputs and deviations after executing the experiment. The examples below are models, not verified live results.
- Check access rights before sharing or publishing an app, report, workflow or externally reachable service.